

Problem 26.10

A perpetuity-immediate has annual payments of $1, 3, 5, 7, \dots$. If the present values of the 6th and 7th payments are equal, find the present value of the perpetuity.

Solution

The payment at the end of year n is

$$2n - 1.$$

Therefore, the 6th payment is 11 and the 7th payment is 13. Since their present values are equal,

$$11v^6 = 13v^7.$$

Dividing by $13v^6$ gives

$$v = \frac{11}{13}.$$

Hence,

$$1 + i = \frac{1}{v} = \frac{13}{11},$$

so

$$i = \frac{2}{11}.$$

The perpetuity has first payment $P = 1$ and annual arithmetic increase $Q = 2$. Its present value is

$$PV = \frac{P}{i} + \frac{Q}{i^2}.$$

Substituting $P = 1$, $Q = 2$, and $i = 2/11$,

$$PV = \frac{1}{2/11} + \frac{2}{(2/11)^2}.$$

Thus,

$$PV = \frac{11}{2} + \frac{121}{2} = 66.$$

Therefore, the present value of the perpetuity is

$$\boxed{66}.$$